Budget — AKG-E Full Proposal (DRAFT)

(DRAFT — for PI / lab-admin review; dollar amounts are placeholders)

Project: Agentic Knowledge Graphs for AI-Accelerated Engineering with Simulation-Grounded Validation
PI: Kereshmeh Afsari **Co-PI:** Chris Brown **Student Researcher:** Jason Cusati
Sponsor: Amazon–Virginia Tech Initiative, CFP 2026–2027
Period: 12 months (3-month aggressive build + 9-month dissemination / scaling)

Category	Amount
Personnel	
GRA #1 — Cusati (12 mo, full-time; primary student researcher)	\$50,000
Stipend (12-mo full-time GRA)	\$30,000
Tuition (annual VT graduate, sponsored-research rate)	\$18,000
Benefits / fringe (~7% on stipend)	\$2,000
GRA #2 — TBD (12 mo, ~25% effort; engineering support)	\$14,000
Stipend (pro-rated)	\$8,000
Tuition (pro-rated)	\$5,000
Benefits / fringe	\$1,000
Faculty summer salary — Afsari, 1 mo (optional)	\$14,000
Materials and Computing	
LLM API tokens (development + benchmark runs)	\$8,000
AWS Bedrock + cloud compute (KG hosting, serving, scaling experiments)	\$10,000
Simulation software / FEA licenses (FEniCS open-source + utilities)	\$1,000
Travel	
Conference travel (AI-for-engineering / agentic AI venue)	\$2,000
Amazon collaboration travel (Seattle / Arlington)	\$1,000
Total Direct Costs	\$100,000
Indirect Costs (gift research — no F&A applied)	\$0
Total Project Cost	**\$100,000**

Justification — LLM API Tokens. AKG-E is, by construction, a high-volume LLM workload: each iteration of the propose → VVUQ → feedback loop spends tokens on retrieval-augmented generation over the curated knowledge graph, on structured graph queries that surface code provisions and physical-law constraints, on simulation feedback ingestion (FEA / digital-twin / PDE solver outputs returned to the agent), and on the refinement step that consumes gate-emitted violations to produce the next candidate artifact. The benchmark evaluation amplifies this footprint substantially: three baselines (B0 Claude alone; B1 Claude + graph-RAG; B2 full AKG-E) are run across three engineering domains (construction, transportation, energy), with $n \geq 5$ random seeds per cell and bootstrap resampling for 95% confidence intervals on the headline correctness, calibration, and efficiency metrics. The full benchmark therefore drives thousands of multi-turn agent runs at production-quality model tokens, and the LLM API line is sized to support both day-to-day development and full benchmark execution without throttling. Separating this line from cloud compute prevents the token spend from being absorbed into a generic AWS line item where Amazon reviewers and VT lab admin cannot audit it independently.

Notes for reviewers: Total ask of \$100,000 matches the Amazon-VT CFP guidance ("~\$100K range, funded as gift research"). The award is gift research, so no F&A indirect cost is applied; if Amazon elects to rescope as sponsored research, F&A will be added at the VT-applicable rate and the personnel lines adjusted accordingly. Personnel allocation reflects that GRA #1 (Cusati) is the primary student researcher executing the build, with GRA #2 providing supplementary engineering bandwidth at ~25% effort during peak build phases. Dollar amounts are draft estimates pending PI sign-off; final GRA stipend, tuition, and fringe rates for the 2026–2027 academic year are subject to VT lab-admin confirmation. The faculty summer salary line is optional and may be zeroed at the PI's discretion (reducing total to \$86,000, with the difference reallocatable to compute / LLM tokens if Amazon prefers a full \$100K ask).